Enroll. No._____

MARWADI UNIVERSITY
FOET, MU-FOT
INFORMATION AND COMMUNICATION TECHNOLOGY
MU FINAL EXAM
SEMESTER 5 - Winter - Dec-2025

Subject : COMPUTER NETWORKS (01CT0503)

Date : 09-Dec-2025 **Time : 2 Hours** **Total Marks : 60**

Instructions :

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Do not write/sign/indication/tick mark anything other than Enroll No. at a specific place on the question paper.**

		Marks	BL	CO
Que.1				
(A)	Compare bit stuffing with byte stuffing framing structure.	[4]	L4	CO2

Ans.

Here's a clear 4-mark comparison of bit stuffing vs. byte stuffing framing structures:

Bit Stuffing

Framing structure: Special bit pattern (flag) (e.g., 01111110) marks the start and end of a frame.

Stuffing rule: If sender data contains five consecutive 1s, an extra 0 bit is inserted after them.

At receiver: Removes the stuffed 0s when five consecutive 1s are followed by 0.

Use case: Works at bit-oriented protocols (e.g., HDLC).

Byte Stuffing

Framing structure: Special byte (flag byte) (e.g., DLE STX for start, DLE ETX for end) marks the boundaries.

Stuffing rule: If the flag byte appears in the data, sender adds an escape byte (DLE) before it.

At receiver: Removes the escape byte to recover the original data.

Use case: Works at byte-oriented protocols (e.g., PPP, BISYNC).

✅ In short (tabular form for exam):

Aspect	Bit Stuffing	Byte Stuffing
Unit of framing	Bit pattern (e.g., 01111110)	Byte (e.g., DLE STX / DLE ETX)
Stuffing method	Insert 0 after five consecutive 1s	Insert escape byte before flag
Receiver action	Removes extra 0s	Removes escape byte
Protocol type	Bit-oriented (HDLC)	Byte-oriented (PPP, BISYNC)

Bit Stuffing (Bit-Oriented)

```
----------------------------
| Flag | Data with stuffing | Flag |
 01111110   0111110 0 1 00101   01111110
```

Explanation:
- Flag = 01111110
- After 5 consecutive 1s, a 0 is stuffed (shown above).
- Receiver removes the stuffed 0.

Byte Stuffing (Byte-Oriented)

```
-----------------------------
| DLE STX | Data with stuffing | DLE ETX |
Start      A  DLE → DLE DLE  B    End
```

Explanation:
- Flag bytes: DLE STX (start), DLE ETX (end)
- If a DLE appears in data, an extra DLE is inserted (escaped).
- Receiver removes the extra DLE.

(B)	Justify why subnetting is important in a corporate network. Provide three practical reasons with examples.	[6]	L4	CO6
Ans.	Efficient IP Address Utilization: Prevents wasting IP addresses by assigning only as many addresses as needed per department. Example: HR needs 30 hosts → subnet with 32 addresses instead of using a /24. Improved Network Performance: Reduces broadcast domains, minimizing unnecessary traffic. Example: Each department has its own subnet to reduce collisions. Enhanced Security & Management: Allows isolation of different departments or networks for policy enforcement. Example: Sales subnet is separated from IT subnet, restricting access between them.			
Que.2				
(A)	Apply the classification of networks based on scale to propose a communication setup for a smart city. Explain how each network type (PAN, LAN, MAN, WAN) fits into your proposal.	[8]	L3	CO1
Ans.	Designing a communication setup for a smart city requires integrating various types of networks—PAN, LAN, MAN, and WAN—to support a range of services like smart homes, transportation, healthcare, utilities, and governance. 1. Personal Area Network (PAN): Use Case in Smart City: Wearable health monitoring devices (e.g., smartwatches). Bluetooth-based keyless entry systems in smart homes and cars. RFID-enabled systems for public transit cards or inventory tracking. Role in Communication: PANs serve individual users or devices with short-range, low-power connectivity. They are essential for user-specific, localized communication in smart devices and homes. 2. Local Area Network (LAN): Use Case in Smart City: Networking within government buildings, hospitals, or police departments. IoT sensors within smart buildings for HVAC, lighting, and security systems. Local servers and employee workstations in municipal offices. Role in Communication: LANs provide high-speed communication over a restricted area (e.g., a building or campus), enabling fast data exchange among smart systems within a local environment.			

3. Metropolitan Area Network (MAN):
Use Case in Smart City:

Connecting various buildings like schools, administrative offices, libraries, and surveillance networks across the city.
City-wide Wi-Fi or fiber infrastructure enabling broadband for citizens and government services.
Traffic control systems using MAN to interlink road sensors and control centers.

Role in Communication:
MANs bridge LANs across a city, offering medium-range connectivity and ensuring city-wide data sharing for coordinated operations and monitoring.

4. Wide Area Network (WAN):
Use Case in Smart City:

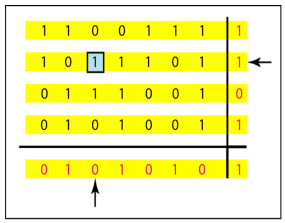
Interconnection between multiple smart cities or rural-urban hubs.
Cloud-based data centers and national e-governance portals.
Secure connectivity to global services like weather forecasting, navigation, and emergency services.

Role in Communication:
WANs provide long-distance communication, ensuring that the smart city can interact with remote servers, national agencies, and global platforms securely and reliably.

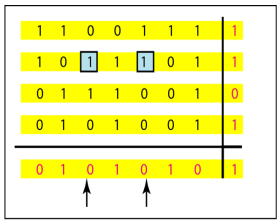
(B)	Compare odd parity and even parity check. Analyze how 2 dimension parity check helps to detect the errors.	[4]	L4	CO2

Ans.

Feature	Even Parity	Odd Parity
Definition	Adds a parity bit so total number of 1s becomes even	Adds a parity bit so total number of 1s becomes odd
Purpose	Ensures even count of 1s in transmitted data	Ensures odd count of 1s in transmitted data
Error Detection	Detects any single-bit error by causing parity mismatch	Detects any single-bit error similarly
Usage	Common in communication systems and memory checks	Used when odd-parity hardware is implemented



b. One error affects two parities



c. Two errors affect two parities

OR

(A)	Analyze the differences between User EXEC, Privileged EXEC, and Global Configuration modes in terms of their purpose and command availability.	[8]	L4	CO1

Ans.

Cisco IOS divides CLI access into several modes, each designed with different levels of permission and command functionality. The three primary modes are User EXEC, Privileged EXEC, and Global Configuration Mode.

1. User EXEC Mode

Prompt: Router>

Purpose:
This is the initial mode entered after accessing the device. It provides limited access for basic monitoring and diagnostics.

Command Availability:

Basic commands like ping, traceroute, and show (some limited).

No access to configuration or system-level changes.

Example:

Router> show version

2. Privileged EXEC Mode

Prompt: Router#

Purpose:
This mode allows complete access to monitoring commands and is the gateway to configuration modes.

Command Availability:

All show commands (e.g., show running-config, show interfaces)

Access to copy, reload, debug, erase, etc.

Entry into global configuration mode using configure terminal.

Example:

Router# show running-config

3. Global Configuration Mode

Prompt: Router(config)#

Purpose:
Used to make system-wide configuration changes to the device.

Command Availability:

Commands to set hostname, passwords, interfaces, routing protocols, banners, etc.

Access to sub-modes like interface or line configuration.

Example:

Router(config)# hostname CoreRouter

Comparison Table

Aspect	User EXEC Mode	Privileged EXEC Mode	Global Configuration Mode
Prompt	Router>	Router#	Router(config)#
Access Level	Limited (monitoring only)	Full monitoring + control	Device configuration
Can configure device?	❌	❌	✅
Typical Commands	ping, show	show, copy, debug	hostname, interface, etc.

(B)	Compare HDLC with PPP protocol.	[4]	L4	CO2

Ans. Here's a concise 4-mark comparison of HDLC vs PPP protocol:

Feature	HDLC (High-Level Data Link Control)	PPP (Point-to-Point Protocol)
Type	Bit-oriented protocol	Byte-oriented protocol
Standard Use	Mainly used in synchronous networks (e.g., WANs, leased lines)	Used in point-to-point links, often over serial/modem connections
Frame Structure	Uses flags, addresses, control fields	Simpler frame with flag, address, control = fixed

Feature	HDLC (High-Level Data Link Control)	PPP (Point-to-Point Protocol)
Protocol Support	Supports only synchronous, single protocol	Supports multiple protocols (IP, IPv6, etc.) via protocol field
Complexity	More complex, supports modes (NRM, ABM)	Simpler and more flexible

✅ Summary:

HDLC = bit-oriented, protocol-specific, synchronous.

PPP = byte-oriented, multiprotocol, works over point-to-point links.

Que.3				
(A)	Define multimedia and explain its importance. Support your answer with memory retention statistics (20%–80%) and list any four benefits of using multimedia.	[8]	L4	CO5

Ans.

Definition of Multimedia

Multimedia is the integrated use of multiple media elements — text, graphics/images, sound/audio, animation, and video — in a single application to create a rich, engaging, and interactive experience. In web-based context, it refers to online content that combines two or more of these elements with user interactivity.

Importance of Multimedia – Supported by Memory Retention Research

Multimedia is extremely important in education, training, business, and entertainment because human beings learn and remember far better when multiple senses are engaged and when they actively participate. Research proves the following retention rates:

Method of Receiving Information	Approximate Retention Rate
Only seeing (passive reading/looking)	20%
Only hearing (lectures, audio)	30%
Seeing + Hearing together (video, presentation)	50%
Seeing + Hearing + Active Interaction (clicking, choosing, doing)	80%

This dramatic jump from 20–30% to 80% clearly shows why multimedia (especially interactive multimedia) is so powerful — it moves learners from passive recipients to active participants, which creates deeper understanding and long-term memory.

Four Key Benefits of Using Multimedia

Improves Retention and Learning – By combining visual, auditory, and interactive elements, users remember up to 80% of the content (as shown in the table above).

Addresses Multiple Learning Styles – Some people learn best by reading (text), some by watching (video/animation), some by listening (audio), and some by doing (interaction). Multimedia supports all types simultaneously.

Enhances User Engagement and Enjoyment – Dull facts become interesting when presented with animations, music, videos, and attractive graphics (e.g., a boring history lesson becomes exciting with a video reconstruction).

Allows User Control and Personalization – Through interactivity and non-linear navigation, users decide what to see, when, and how deeply — making the experience more effective and satisfying (e.g., clicking topics in an e-learning portal).

(B)	What is Jitter? Is jitter depending on congestion? Justify your answer.	[6]	L4	CO5

Ans.

Yes, jitter depends on congestion.

Justification

1. What is Jitter?

Jitter is the variation in packet delay (i.e., packets do not arrive at equal time intervals).
It is especially important in real-time applications like:

Voice over IP (VoIP)
Video conferencing
Online gaming
2. How Congestion Causes Jitter
(a) Congestion increases queueing delay
When the network becomes congested:
Routers' buffers start filling up
Some packets wait longer in queues
Some packets experience shorter delay
Thus, delay becomes inconsistent, leading to high jitter.
(b) Packet drops and retransmissions
Congestion leads to:
Packet loss
Retransmissions
Variable end-to-end delay
This creates irregular arrival times, increasing jitter further.
(c) Sharing of bandwidth
When many flows compete for router bandwidth:
Some packets get served sooner
Others wait longer
This uneven servicing produces delay variation, i.e., jitter.
Conclusion
Yes, jitter depends heavily on congestion.
As congestion increases, queueing delays become unpredictable, packet loss rises, and transmission times vary—leading to higher jitter.

OR

(A)	(a) Explain the hierarchical structure of DNS name space with a diagram. (b) What are Generic and Country top-level domains? Give 2 examples each. (c) Who controls subdomain allocation?	[8]	L3	CO5

Ans. (a) Explain the hierarchical structure of DNS name space with a diagram.

DNS (Domain Name System) uses a hierarchical inverted-tree structure to translate domain names into IP addresses.
The hierarchy consists of multiple levels:
Root Level
Represented by a dot (.).
Highest level in the DNS structure.
Manages pointers to all Top-Level Domains (TLDs).

Top-Level Domains (TLDs)
Directly under the root.
Examples: .com, .org, .edu, .in, .uk

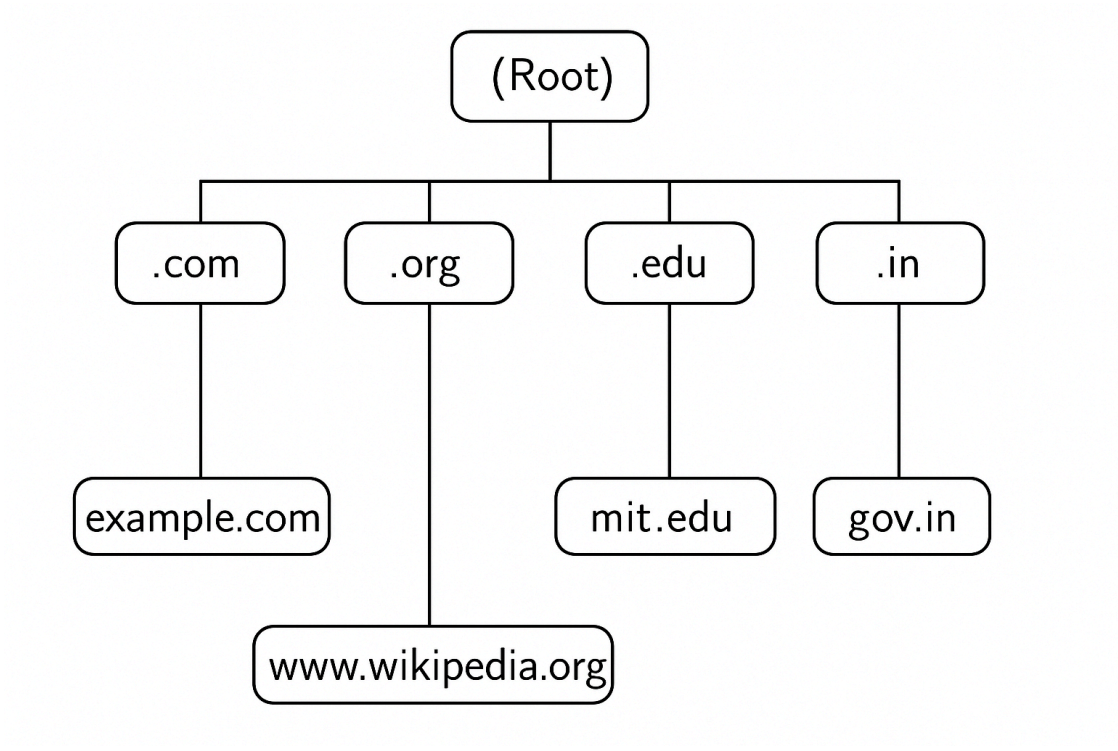
Second-Level Domains
Domain names registered by organizations/individuals.

Example: example in *example.com*

Subdomains
Created by organizations for internal structure.
Example: www.example.com, cs.mu.ac.in

Host (individual machine)
Final leaf node, representing an actual device.

Diagram of DNS Hierarchy (Simple Text Diagram)



(b) What are Generic and Country top-level domains? Give 2 examples each.
1. Generic Top-Level Domains (gTLDs)
These are global domains not tied to any particular country.
They are generally meant for organizations, businesses, or purposes.
Examples:
.com – Commercial
.org – Organizations
(Other examples: .edu, .net, .gov)
2. Country Code Top-Level Domains (ccTLDs)
These represent specific countries and use two-letter ISO country codes.
Examples:
.in – India
.uk – United Kingdom
(Other examples: .us, .jp, .au)

(c) Who controls subdomain allocation?
Subdomain allocation is controlled by the administrator/owner of the parent domain.
Example:
If an organization owns mu.ac.in, then the MU domain administrator can create subdomains like:

	cs.mu.ac.in exam.mu.ac.in admission.mu.ac.in They have full authority to create, modify, or remove subdomains.			
(B)	With a neat diagram, explain the 10-step DNS resolution process for a remote name lookup.	[6]	L3	CO5

Ans.

When a client resolves a remote domain (e.g., www.example.com, DNS follows a hierarchical, iterative process across multiple name servers.

10-Step DNS Resolution Process

User enters the domain
The browser receives a request for www.example.com.

Check browser cache
The browser first checks its own DNS cache for previously resolved entries.

Check OS resolver cache
If not found, the operating system resolver checks local cache (e.g., /etc/resolv.conf or Windows DNS cache).

Query sent to Recursive Resolver (ISP DNS / Public DNS)
The client forwards the DNS query to its configured recursive resolver (e.g., 8.8.8.8).

Recursive resolver checks its own cache
If no cached answer exists, it starts a full DNS lookup.

Query to Root Name Server
The resolver asks a root server: “Where is the .com TLD server?”
Root replies with the list of .com TLD servers.

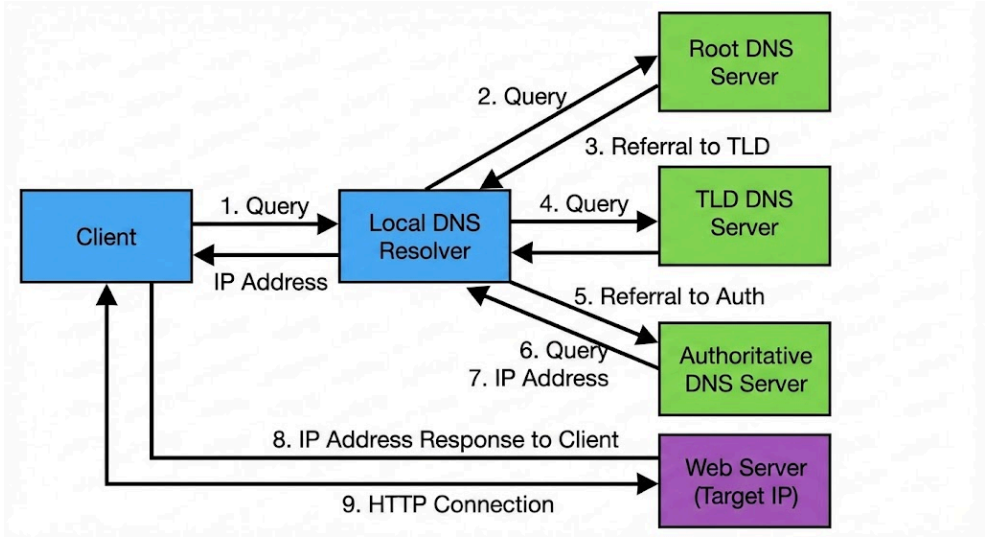
Query to TLD (.com) Name Server
The resolver asks the .com server: “Where is the authoritative server for example.com?”
TLD replies with the authoritative servers for example.com.

Query to Authoritative Name Server
The resolver queries the authoritative server:
“What is the IP of www.example.com?”
It responds with the A record (IPv4 address).

Recursive resolver returns the final answer
The resolver sends the IP address back to the client machine.

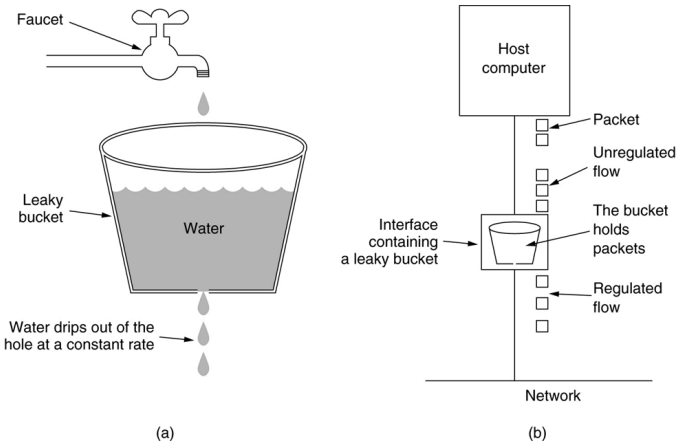
Client stores result in cache and connects to server
The browser caches the response (for TTL duration) and makes a TCP/HTTPS connection to the resolved IP.

Sample Diagram:



Que.4				
(A)	Sketch and explain leaky bucket algorithm.	[6]	L4	CO5

Ans.



The Leaky Bucket Algorithm is a traffic shaping and traffic policing technique used in computer networks to regulate the flow of packets into the network.

It ensures that data is transmitted at a constant, steady rate, regardless of how bursty the incoming traffic is.

Basic Concept

Incoming packets enter a bucket (a buffer).

The bucket leaks data out at a fixed, constant rate, irrespective of how fast the input arrives.

If the bucket becomes full, extra incoming packets are discarded.

Working of the Algorithm

Packets arrive at the bucket (which acts as a queue).

The bucket has a fixed capacity (buffer size).

Packets are transmitted one by one at a constant rate (e.g., 1 packet per millisecond).

If incoming traffic is faster than the drain rate:

The bucket fills up.

Once full, any additional packets are dropped.

Thus, bursty input traffic becomes smooth output traffic.

Characteristics

Output rate is strictly constant, even if input is bursty.

Prevents network congestion by maintaining stable load.

Does not allow bursts during transmission (unlike token bucket).

Advantages

Simple and easy to implement.
Ensures smooth and predictable network traffic.
Prevents sudden bursts that may congest routers.
Disadvantages
Does not allow bursts even if the network is free.
Possible packet drops if incoming rate is high and bucket is small.
Less flexible compared to the Token Bucket Algorithm.

(B)	Compare Go Back to N with Selective Repeat reQuest protocol.	[4]	L4	CO2

Ans. Here's a concise 4-mark comparison of Go-Back-N (GBN) vs Selective Repeat (SR) ARQ:

Comparison Table (Exam-friendly)

Feature	Go-Back-N (GBN)	Selective Repeat (SR)
Window size	Sender: N, Receiver: 1	Sender: N, Receiver: N
Acknowledgment	Cumulative ACK (acknowledges all previous frames)	Individual ACK for each frame
Error handling	On error, retransmit all frames from error onwards	Retransmit only the erroneous frames
Efficiency	Less efficient under high error rate (many frames retransmitted)	More efficient, avoids unnecessary retransmission
Receiver buffer	Needs minimal buffer (only next expected frame)	Needs buffer for out-of-order frames

Short summary (1–2 lines)

GBN: Simple, retransmits multiple frames after error, low buffer requirement.

SR: Complex, retransmits only erroneous frames, higher efficiency and buffer requirement.

OR

(A)	Discover the role of SMTP in email transfer. Draw and label the envelope vs message structure.	[6]	L3	CO5

Ans. Role of SMTP in Email Transfer

SMTP (Simple Mail Transfer Protocol) is an application-layer protocol used for sending, routing, and forwarding emails across the Internet. It mainly handles push-based email delivery from the sender to the recipient's mail server.

Functions of SMTP:

Mail Submission:
SMTP transfers email from the Mail User Agent (MUA) to the Mail Transfer Agent (MTA) (e.g., Outlook → Gmail server).

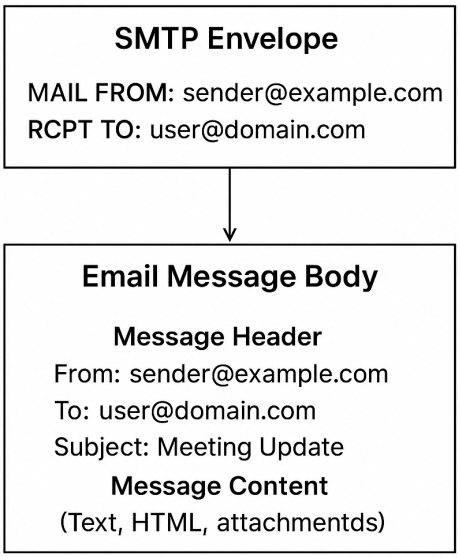
Mail Relay:
SMTP servers forward the email across multiple mail servers until it reaches the recipient's domain.

Mail Delivery:
The recipient's MTA receives the mail and hands it to a Mail Delivery Agent (MDA).

Commands & Replies:
SMTP uses commands like HELO, MAIL FROM, RCPT TO, DATA, and replies like 250 OK.

Text-Based Protocol:
SMTP sends emails as plain text over TCP, usually on port 25 / 587 / 465.

Store-and-Forward Mechanism:
If the receiver is offline, SMTP queues the email and retries delivery.



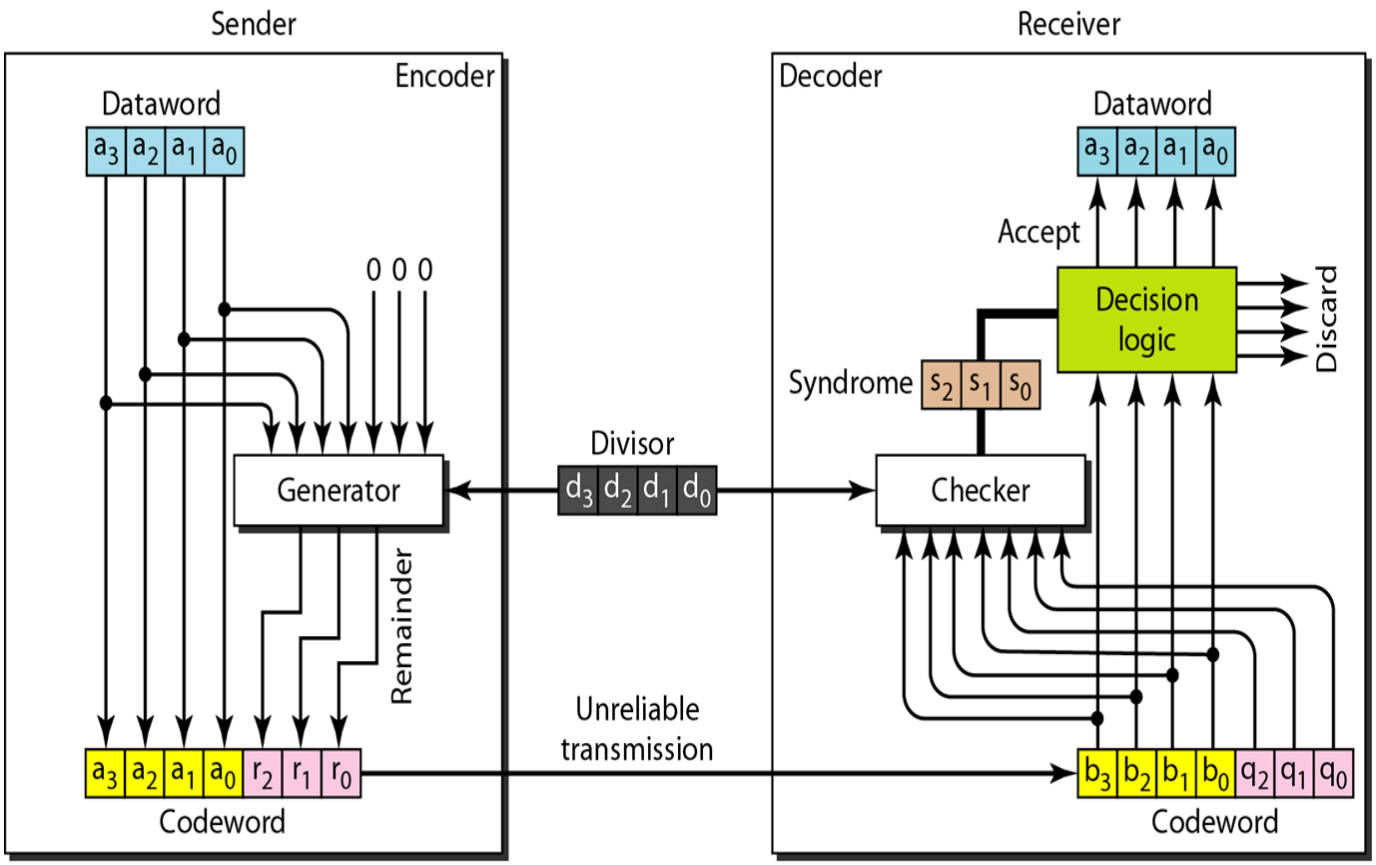
Difference:

Envelope:
Used by SMTP for routing and delivery (invisible to the user). Contains MAIL FROM and RCPT TO.

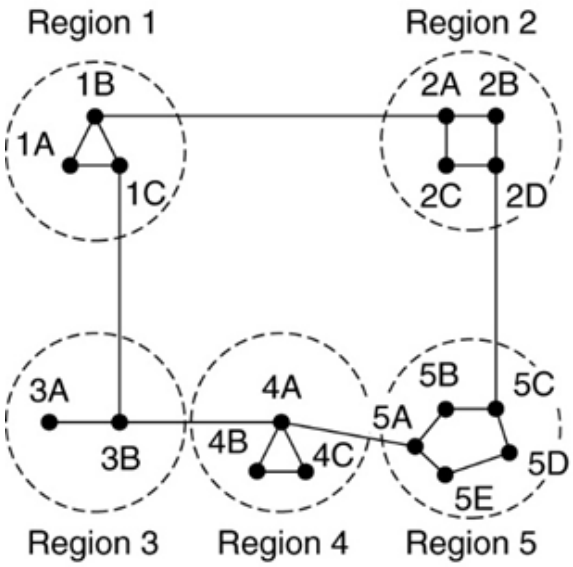
Message:
The actual email content, containing headers (From, To, Subject) and body (text, attachments).

(B)	Analyze the CRC encoding and decoding process with block diagram.	[4]	L4	CO2

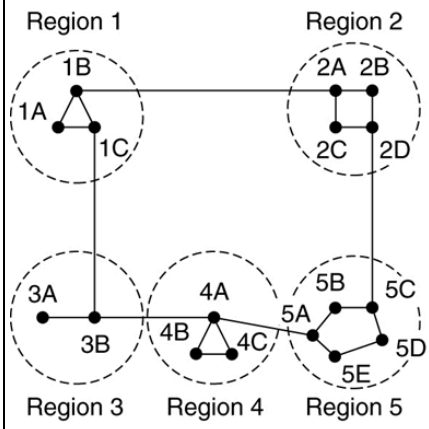
Ans.



Que.5

(A)	What is the need of Hierarchical routing algorithm? Build the full routing table and Hierarchical routing table for node 1A for the below given network. 	[8]	L4	CO6

Ans.



(a)

Full table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

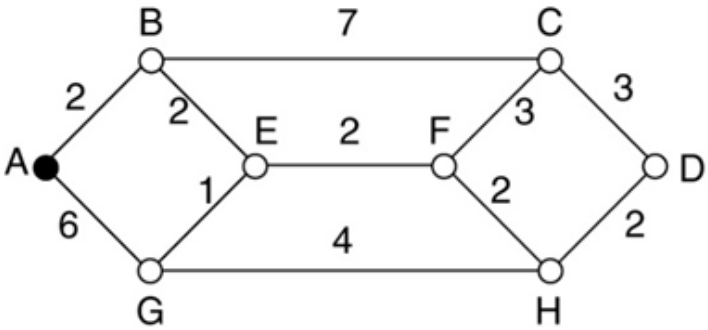
Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

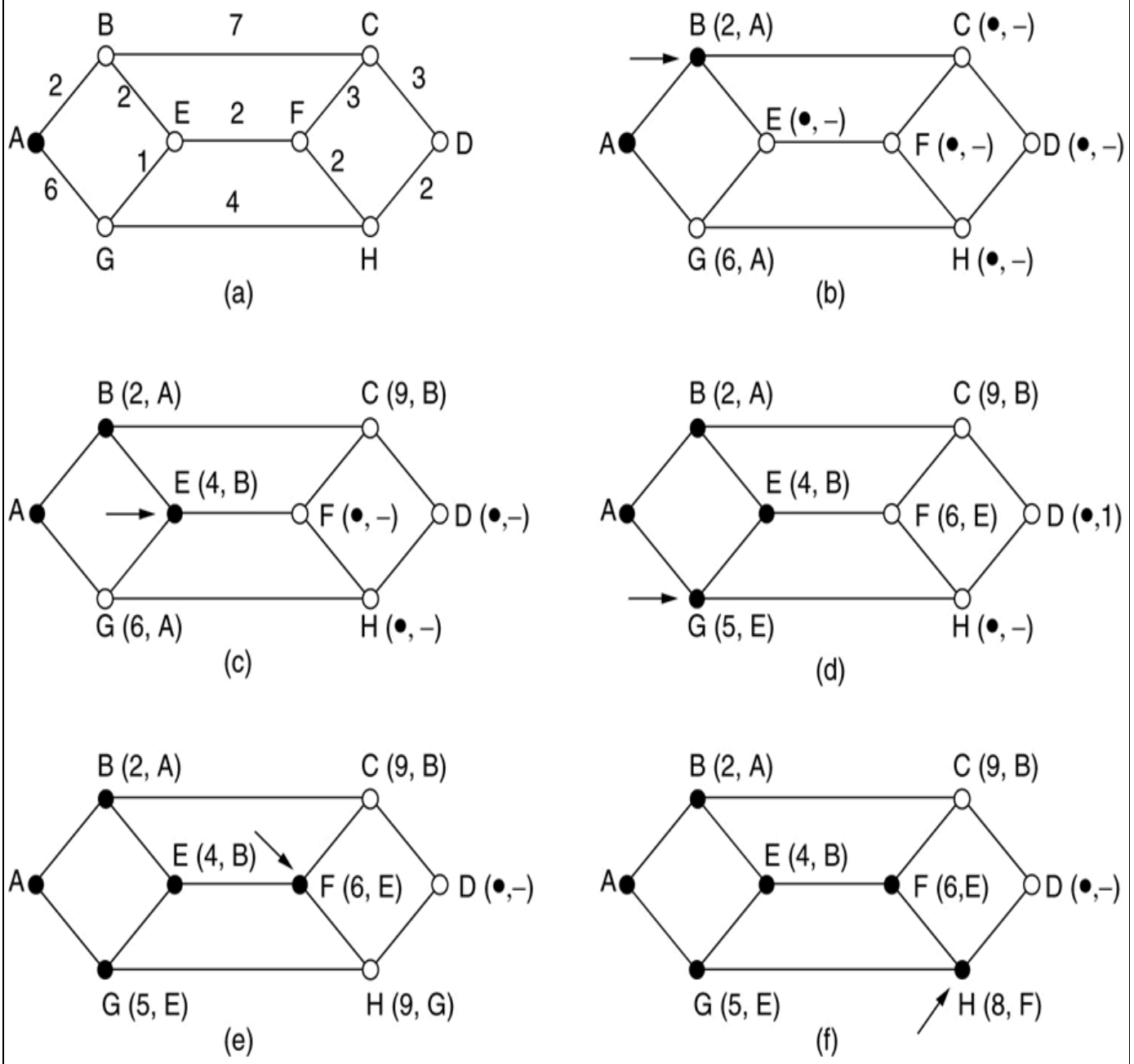
(B)	Draw a simple topology of two PCs connected via a switch and a router. Label all the cables used and explain why each cable type is chosen.	[6]	L3	CO6

Ans. Topology Diagram:

```
PC1 ——(Straight-through)——┐
                          |
                [ Switch ] ——(Straight-through)—— [ Router ]
                          |
PC2 ——(Straight-through)——┘
```

	Explanation of Cable Types: PC to Switch (PC1–Switch & PC2–Switch): Straight-through cable is used because end devices (PCs) connect to networking devices (switch). Switch to Router: Straight-through cable is used because routers and switches are different device types. Reason for Choice: Straight-through cables are used when connecting different device types (PC–Switch, Switch–Router). Crossover cables are only needed for similar devices (e.g., Switch–Switch, Router–Router) in older setups without auto-MDIX.			
	OR			
(A)	Analyze below given subnet and identify the shortest path from node A to node D using shortesh path routing algorithm. Mention all working nodes and calculation.  	[8]	L4	CO6

Ans.



(B)	In a lab setup, you have the option to connect two Cisco switches using either a crossover cable or a fiber optic cable. Evaluate both options and recommend the better choice for (a) a classroom lab and (b) a data center environment.	[6]	L3	CO6

Ans.

Crossover Cable (UTP Copper):

Advantages:

Low cost and easy availability.

Simple to set up for short distances.

Suitable for basic lab practice.

Disadvantages:

	Limited bandwidth (usually up to 1 Gbps for older Cat5e, 10 Gbps for Cat6/6a). Limited distance (up to 100 meters). More prone to electromagnetic interference. Fiber Optic Cable: Advantages: Very high bandwidth (10 Gbps to 400 Gbps). Supports long-distance connections (hundreds of meters to kilometers). Immune to electrical interference. Disadvantages: Higher cost for both cable and transceivers (SFP modules). Requires careful handling and skilled installation. Recommendation: (a) Classroom Lab: Use a crossover cable because it is inexpensive, easy to connect, and sufficient for small lab environments where distance and bandwidth requirements are minimal. (b) Data Center Environment: Use fiber optic cable because it supports high speed, longer distances, and reliable performance in enterprise-level networks.

---Best of Luck---

MARWADI UNIVERSITY
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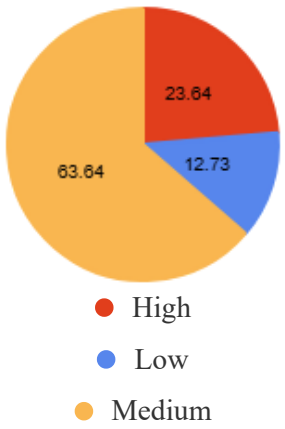
INFORMATION AND COMMUNICATION TECHNOLOGY
Semester SEMESTER 5 - Winter

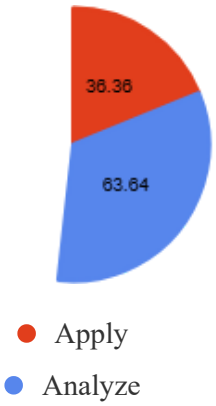
Subject : COMPUTER NETWORKS (01CT0503)
Date : 09-Dec-2025 **Time : 2 Hours** **Total Marks : 60**

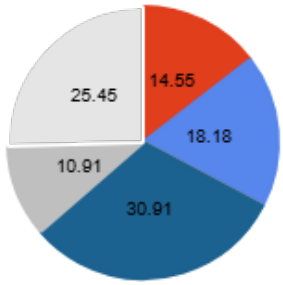
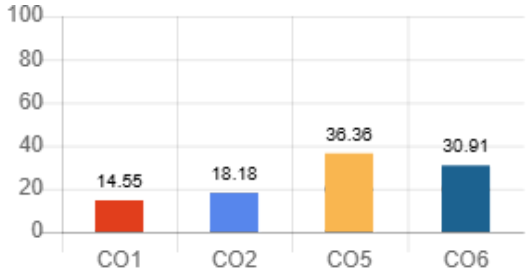
Difficulty Level	Weightage	No of Question	Total Marks	Question List
High	23.64	5	26	2(B), 4(A), 4(B), 5(A)
Low	12.73	2	14	2(A), 5(B)
Medium	63.64	11	70	1(A), 1(B), 2(A), 2(B), 3(A), 3(B), 4(A), 5(A), 5(B)

Module Name	Weightage	No of Question	Total Marks	Question List
Unit 1: Introduction:	14.55	2	16	2(A)
Unit 2: Data Link Layer	18.18	5	20	1(A), 2(B), 4(B)
Unit 3: Medium Access Control Sub layer	0.00	0	0	
Unit 4: Network Layer	30.91	5	34	1(B), 5(A), 5(B)
Unit 5: Transport Layer	10.91	2	12	3(B), 4(A)
Unit 6: Application Layer	25.45	4	28	3(A), 3(B), 4(A)

Blooms Taxonomy	Weightage	No of Question	Total Marks	Question List
L1 - Remember / Knowledge	0.00	0	0	
L2 - Understand	0.00	0	0	
L3 - Apply	36.36	6	40	2(A), 3(A), 3(B), 4(A), 5(B)
L4 - Analyze	63.64	12	70	1(A), 1(B), 2(A), 2(B), 3(A), 3(B), 4(A), 4(B), 5(A)
L5 - Evaluate	0.00	0	0	
L6 - Higher order Thinking / Creative	0.00	0	0	







- Unit 1: Introduction:
- Unit 2: Data Link Layer
- Unit 3: Medium Access Control Sub layer
- Unit 4: Network Layer
- Unit 5: Transport Layer
- Unit 6: Application Layer